Education

Sun Yat-sen University, Guangzhou, China 2022 – 2027 (expected)

PH.D. in Computer Science (CS)

Xidian University, Shaanxi, China 2018 – 2022

B.S. in Computer Science (CS)

Skills

- Open-Source Contributor: **[gLLM]**, **vLLM**, **SGLang** (Implemented Encoder-Prefill-Decode (EPD) Disaggregation for multimodal LLMs, enabling elastic encoder scaling and significantly improving TTFT for vision-heavy workloads [PR #12263], [LMSYS Blog])

Publications

- **arXiv'25 Preprint**
  Tianyu Guo, Tianming Xu, Xianjie Chen, Junru Chen, Nong Xiao, Xianwei Zhang, "RServe: Overlapping Encoding and Prefill for Efficient LMM Inference", arXiv:2509.24381, 2025.
- **NeurIPS'25 (CCF-A)**
  Hongxin Xu*, **Tianyu Guo*** and Xianwei Zhang, "DynaPipe: Dynamic Layer Redistribution for Efficient Serving of LLMs with Pipeline Parallelism", The Thirty-Ninth Annual Conference on Neural Information Processing Systems, San Diego, CA, United States, December.
- **SC'25 (CCF-A)**
  Tianyu Guo, Xianwei Zhang, Jiangsu Du, Zhiguang Chen, Nong Xiao, Yutong Lu, "gLLM: Global Balanced Pipeline Parallelism System for Distributed LLM Serving with Token Throttling", The International Conference for High Performance Computing, Networking, Storage, and Analysis, St. louis, MO, United States, November.
- **Euro-Par'25 (CCF-B)**
  Tianyu Guo, Hande Dong, Yichong Leng, Feng Liu, Cheater Lin, Nong Xiao and Xianwei Zhang, "EFIM: Efficient Serving of LLMs for Infilling Tasks with Improved KV Cache Reuse", The 31st International European Conference on Parallel and Distributed Processing, Dresden, Germany, August.
- **ASP-DAC'25 (CCF-C)**
  Mengyue Xi, **Tianyu Guo**, Xuanteng Huang, Zejia Lin, Xianwei Zhang, "Mpache: Interaction Aware Multi-level Cache Bypassing on GPUs", The 30th Asia and South Pacific Design Automation Conference, Tokyo Odaiba Miraikan, Japan, January.
- **DAC'24 (CCF-A)**
  Tianyu Guo, Xuanteng Huang, Kan Wu, Xianwei Zhang and Nong Xiao, "SMILE: LLC-based Shared Memory Expansion to Improve GPU Thread Level Parallelism", The 61st ACM/IEEE Design Automation Conference, San Francisco, CA, United States, June.

Experience and Projects

Scheduling for RL Training (Tencent Hunyuan Intern) 2026

Integrated RL training pipeline with vLLM router to enable globally coordinated scheduling for large-scale RL training. Implemented KV-cache-aware request routing and scheduling policies to reduce recomputation overhead and improve training throughput, with specific optimizations for Code Agent training scenarios. Extended vLLM router to support Prefill-Decode (PD) disaggregation integration, enabling efficient resource isolation.

gLLM: Distributed LLM Serving System 2026

Designed and iterated a high-performance distributed LLM serving runtime for efficient multi-request inference. Improved end-to-end scheduling and execution efficiency through overlap dispatch and delta scheduling,

and optimized critical runtime paths including communication, memcpy/kernel execution, and CPU overhead. Expanded support for latest dense/MoE and vision-language models (e.g., Qwen3.5, Qwen3-VL), improving throughput-latency trade-offs and production robustness.

REDServe: Overlapping Encoding and Prefill for LMM Inference (RedNote Hilab Intern) 2025

Large multimodal model (LMM) serving systems often suffer from inefficient resource utilization due to limited overlap between multimodal encoding and language model inference. To address this, we propose REDServe, a serving system that overlaps encoding and prefill execution through fine-grained pipeline scheduling and token-budget coordination. By reducing pipeline idle time and improving workload balance, REDServe significantly enhances GPU utilization and serving efficiency. Experiments show that REDServe achieves up to 109% higher throughput and 66% lower latency compared with existing state-of-the-art systems.

gLLM: Global Balanced Pipeline Parallelism System with Token Throttling 2025

Pipeline parallelism in LLMs often suffers from inefficiencies due to load imbalance, particularly inter-batch imbalance, which existing methods frequently overlook. To address this, we propose Token Throttling, a dynamic strategy that adapts the number of prefill and decode tokens per batch based on real-time system conditions. By reducing computational imbalance between batches, our approach significantly improves GPU utilization. Experiments show that gLLM outperforms vLLM and SGLang, achieving 11% to 398% higher throughput while maintaining lower latency.

EFIM: Efficient Serving of Code Completion LLMs (Tencent Code Buddy Intern) 2024

Code completion LLMs tend to own the ability of fill-in-the-middle (FIM). We propose EFIM, a transformed prompt format of FIM to unleash the performance potential of KV cache reuse. We also propose an enhanced training procedure on data processing for solving subtoken generation problems. EFIM can lower the latency by 52% and improve the throughput by 98% while maintaining the original code completion capability.

♡ Honors and Awards

The Second Prize in ACTIC of A3 track operator implementation and performance optimization	2023
Top 4.2% nationwide | Top 2.9% worldwide in "leetcode contest"	2023
The Second Prize Scholarship in SYSU	2022-2023
The Second Prize Scholarship in XDU	2018-2019, 2019-2020, 2020-2021